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Machine Learning in Finance

Close-to-Open (C2O) Overnight Equity Strategy

Final Submission Report

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250M NET RETURN	250M NET VOL	250M NET SHARPE	MAX DRAWDOWN
7.31%	4.97%	1.445	-10.19%

Strategy: `phase2_g5_05_expanding` • US Large-Cap Equities, 2010–2024 • Development cutoff: 31 December 2024

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1 Executive Summary and Coursework Map

The final strategy is `phase2_g5_05_expanding` : a daily, dollar-neutral, close-to-open long-short strategy using a volatility-scaled overnight-return target with expanding-window transparent feature-weight estimation. The target is:

```
target = overnight_next / (vol20 / sqrt(252))
```

where `overnight_next` is the close-to-next-open return and `vol20` is trailing 20-day close-to-close volatility, annualised and shifted by one trading day. The feature set is unchanged from the original point-in-time panel. The main research change is a risk-scaled label and an expanding training window, not a larger black-box model.

At the main headline AUM of 250M, the final strategy earns **7.31%** net annual return, **4.97%** net volatility, net Sharpe **1.445**, and max drawdown **-10.19%** over 2010–2024 after commission, auction slippage, borrow costs, and a 5% ADV20 participation cap.

We keep weak evidence in the report. The 2024 year is negative. The 2023–2024 holdout Sharpe is lower than validation. We do not have external prime-broker borrow validation. 1B capacity is clearly constrained. 2025–2026 remains held out for marker evaluation.

Coursework Question Map

The brief contains **24 explicit report questions** across Sections 2–6:

Brief Section	Topic	Explicit Questions	Where Answered Here
Section 2	Daily panel questions	6	Data sources, return identity, earnings timing, short-interest lag, universe and stylised fact
Section 3	Capacity-aware universe	4	Thresholds, participation cap, AUM capacity, binding constraints
Section 4	Borrow filtering	4	Borrow proxy, external validation, tiered cost treatment, gross-to-net borrow impact
Section 5	Alpha model	5	Information set, target, model/training, IC, ablation and weak periods
Section 6	Portfolio and costs	5	Basket/weighting/turnover, 50M/250M/1B performance, cost drag, QuantStats, stress windows

Section 7 adds sceptical-marker audit questions. This report answers those by tying look-ahead, capacity, borrow, robustness, and cost claims to frozen output files.

2 Methodology Pipeline

The pipeline is meant to be boring in the right places. It reads the same local data files, builds the same point-in-time panel, uses the same costs and capacity cap, and writes the same outputs every time. The only promoted modelling change is the volatility-scaled target with expanding-window estimation.

Step	Stage	What the Code Does
1	Load and align data	Read prices, all_data, cheapness scores, earnings, short interest, GICS and benchmark files
2	Build point-in-time features	Lag close/high/low, accounting, valuation and short-interest fields; rank features cross-sectionally
3	Define investable universe	Use prior-year top 1000 market-cap names, then apply the $5 price floor, ADV, volatility, earnings and history filters
4	Train alpha by year	For each scored year, use only earlier years under the expanding-window protocol
5	Form daily baskets	Buy top 3% and short bottom 3%, with at least 15 names each side and equal weights before caps
6	Apply costs and capacity	Use commission, auction slippage, borrow tiers and 5% ADV20 per-name participation cap
7	Audit and report	Write performance, positions, IC, ablation, robustness, borrow, capacity and reproduction files

The main implementation lives in `src/c2o_strategy/final_strategy.py`, with reusable helpers under `src/c2o_strategy/`. The formal code-section artefact is the notebook `notebooks/c2o_research_process_and_tests.ipynb`: it runs the research chain from raw coursework data, regenerates model outputs and report evidence, then audits the saved files and submitted tests. The Makefile is only a convenience wrapper, not the required reproduction path.

3 Data, Trading Clock, and Point-in-Time Panel

3.1 Data Source and Window

The pipeline uses the provided coursework data files under `data/`: daily adjusted prices, market cap, volume, GICS information, accounting/valuation variables, earnings calendar, short-interest proxies, regime data, and S&P 500 total return benchmark. The development sample runs from 1 January 2010 to 31 December 2024. No 2025 or 2026 observations are used.

We do not treat the data as one black-box table. The price file defines the tradable clock and all realised returns; the accounting, cheapness, revision, and short-interest files provide slow-moving cross-sectional state variables; the earnings file is used mainly as an information-timing and exclusion control; and the benchmark file is used only for tear-sheet comparison, not for alpha training.

Input Block	Main Fields Used	Research Role	Timing Treatment
Daily prices	Open, high, low, close, adjusted close, volume, market cap, status	Return decomposition, ADV20, volatility, market-cap universe, execution capacity	Returns use adjusted auction prices; close/high/low-derived features are shifted one day
Accounting and valuation panel	Piotroski, margins, leverage, valuation, revisions, short-interest proxies	Fundamental, revision, borrow-stress, and industry-state features	Lagged before ranking; short-interest proxies receive an additional decision lag
Cheapness scores	Value, quality, health, momentum, final score and score velocity	Compact valuation-quality state variables and candidate alpha controls	Lagged one trading day before use
Earnings calendar	Report date, timing flag, strategy trading date	Prevents event-timing leakage and removes a narrow earnings window	AMC announcements are shifted forward; BMO announcements are available by the same close
GICS and benchmark files	Sector/industry tags and S&P 500 total return index	Industry controls and QuantStats benchmark context	Not used to tune future holdout results

The provided adjusted-price panel is treated as the survivorship-adjusted source. The code loads the full price file up to the development cutoff, freezes the eligible universe from prior-year-end market capitalisation, and does not retroactively filter by 2024 survivors. A selected name remains in the panel until its final available trading date; the universe summary reports these mid-year exits instead of deleting them. Corporate-action consistency is handled by scaling open, high, low, and close with the adjusted-close factor; liquidity uses provided volume and adjusted-close dollar volume.

The resulting research panel is large but still intentionally liquid-biased. We reconcile **5,469,968** stock-days, while the annual universe audit has **5,470,423** candidate filter stock-days. After price, ADV20, volatility, earnings-window, history and universe filters, **3,194,145** stock-days remain tradable, or **58.39%** of the candidate filter table. Year-start universe size is stable at 986-990 names, while median year-start market cap rises from **$2.46B** in 2010 to a peak of **$14.16B** in 2022. This matters for interpretation: the model is not asked to exploit micro-cap overnight anomalies; it is tested on a large-cap, capacity-aware panel where execution assumptions are at least plausible.

Data Diagnostic	Value	Interpretation
Development window	2010-01-01 to 2024-12-31	2025-2026 left untouched for marker evaluation
Reconciled stock-days	5,469,968	Return-identity check is performed before modelling
Tradable stock-days after filters	3,194,145 (58.39%)	Liquidity and event filters materially narrow the panel
Year-start universe count	986-990	Close to the intended top-1000 annual large-cap universe
Mid-year exits recorded	0	Not imposed by us; this is what the supplied panel and audit show after provider handling
Ranked point-in-time features	28	Return/reversal, risk/liquidity, fundamentals, revisions, borrow-stress, industry return

There are two data caveats worth making explicit. First, we rely on the coursework provider's adjusted-price and membership history as the authoritative survivorship-bias-free source rather than rebuilding delisting payouts from raw CRSP-like events. Our audit checks that we do not add a survivor filter ourselves, but it cannot independently prove every delisting payout. Second, volume is used as a lagged liquidity proxy for ADV20 and participation caps, not as a return component. We therefore treat capacity results as conservative execution diagnostics, not as exact auction-volume reconstruction.

The trading decision is made before the day-t close. Positions enter at the close and exit at the next open. Features using day-t close, high, low, accounting, or short-interest data are shifted unless they are known before the decision time. Cross-sectional ranks are formed within the same-day available cross-section only.

3.2 Return Reconciliation

The corporate-action-adjusted OHLC panel passes the open-close return identity check. Open, high, low, and close are aligned through the adjusted-close factor before return construction; volume is kept as the liquidity input for adjusted-dollar-volume and ADV20 filters. We verify:

```
(1 + r_overnight) × (1 + r_intraday) − 1 = close_t / close_(t−1) − 1
```

on **5,469,968** stock-days at tolerance `1e-08`. The fail count is **0**, fail fraction is **0.000000**, and the maximum absolute residual is 4.44e-16.

3.3 Earnings Timing and Short-Interest Lag

Earnings observations are mapped to a strategy trading date. A before-market announcement can be known before that day's close; an after-market announcement cannot be used for the same close decision. The pipeline excludes a ±1 trading-day window around the strategy date.

Example	Ticker	Report Date	Timing	Strategy Date	Excluded Decision Dates
AMC example	MOS	2010-01-05	after	2010-01-05	2010-01-04, 2010-01-05, 2010-01-06
BMO example	RPM	2010-01-06	before	2010-01-05	2010-01-04, 2010-01-05, 2010-01-06

Short-interest variables `dsi`, `dtcn`, and `ddtcn` are read from the provided point-in-time panel. We treat the coursework short-interest panel as already reflecting the official publication lag and the extra two-calendar-day vendor availability lag specified in the brief. On top of that, we apply one additional trading-day decision lag before these fields enter alpha features or borrow-tier rules.

Ticker	Source Feature Date	Decision Date	DSI	DTCN	DDTCN
HLT	2015-01-27	2015-01-28	0.004	1.67	-0.25
HLT	2015-01-28	2015-01-29	0.004	1.67	-0.25
HLT	2015-01-29	2015-01-30	0.004	1.67	-0.25

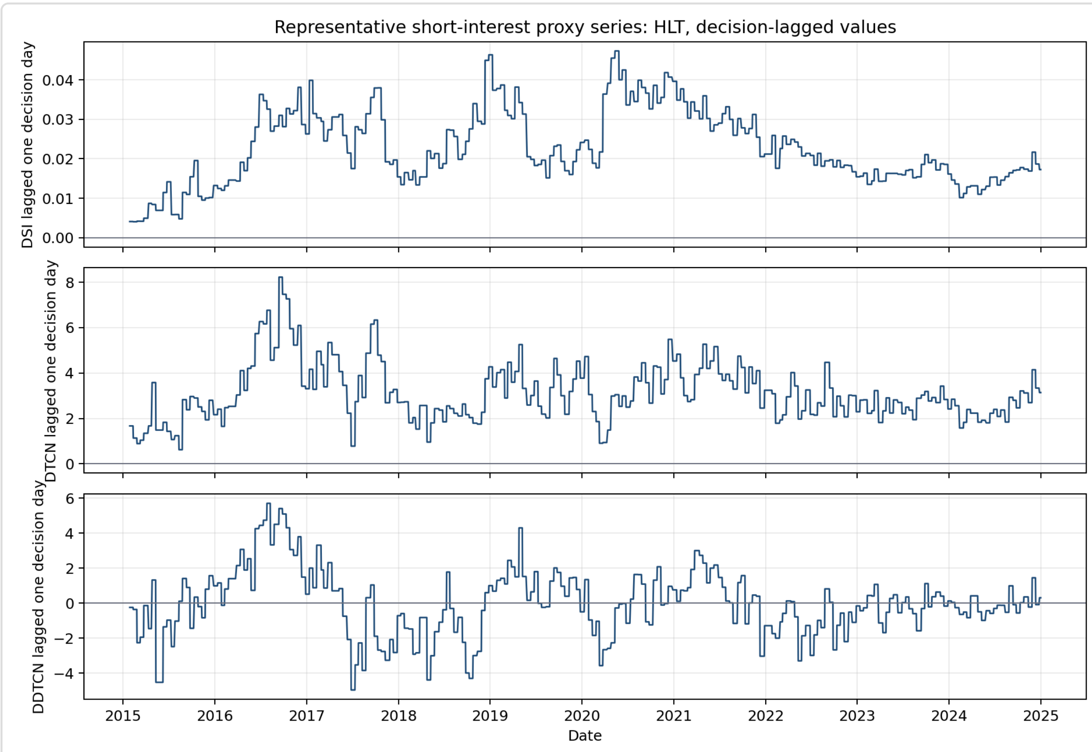

Figure 1. *Representative HLT short-interest proxy series, 2015–2024. Provided point-in-time proxies plus one-trading-day decision lag. No AUM or portfolio cost assumption.*

3.4 Stylised Fact Check

In our eligible equal-weight universe, the result is less clean than the textbook version: the return identity holds exactly, but close-to-close return is larger than overnight return because intraday returns are also positive. We use the overnight effect as motivation, not as proof that any strategy should work. This is one place where we chose not to force the report into the expected story. The stylised fact still matters because it tells us to look carefully at the close-to-open leg. It does not prove that a long-short ML alpha exists. The actual strategy still has to earn its return after costs, borrow and capacity.

The year-by-year decomposition shows meaningful dispersion. In some years (2015, 2016) overnight returns are negative while intraday returns are positive. In other years (2018, 2020, 2022) the overnight stream dominates. This time-variation is why a simple buy-and-hold overnight exposure is not enough and a cross-sectional ranking model is needed to select which stocks to hold overnight.

Across the 15 calendar years, annual overnight-minus-intraday return ranges from -23.30% to 31.01%, with a cross-year standard deviation of 17.28%. Overnight return exceeds intraday return in 7 of 15 years. This mixed evidence is important: the equal-weight large-cap universe supports looking at the overnight leg, but it does not mechanically reproduce the textbook index story every year.

Year	Overnight	Intraday	Close-to-Close	ON - Intraday	Avg Names
2010	9.79%	16.05%	26.98%	-6.26%	986
2011	5.85%	-3.61%	1.76%	9.46%	986
2012	3.93%	16.36%	20.71%	-12.43%	988
2013	13.69%	22.09%	38.55%	-8.39%	988
2014	10.12%	2.41%	12.57%	7.71%	987
2015	-1.52%	0.55%	-1.22%	-2.07%	986
2016	-1.71%	21.59%	19.30%	-23.30%	987
2017	15.13%	4.96%	20.70%	10.17%	988
2018	12.15%	-17.76%	-7.95%	29.91%	988
2019	12.47%	15.52%	29.77%	-3.05%	988
2020	24.74%	-3.04%	20.31%	27.78%	989
2021	29.61%	-1.40%	27.43%	31.01%	990
2022	-10.50%	-2.61%	-13.08%	-7.89%	988
2023	6.41%	12.04%	18.83%	-5.63%	988
2024	21.57%	-5.89%	14.14%	27.45%	989

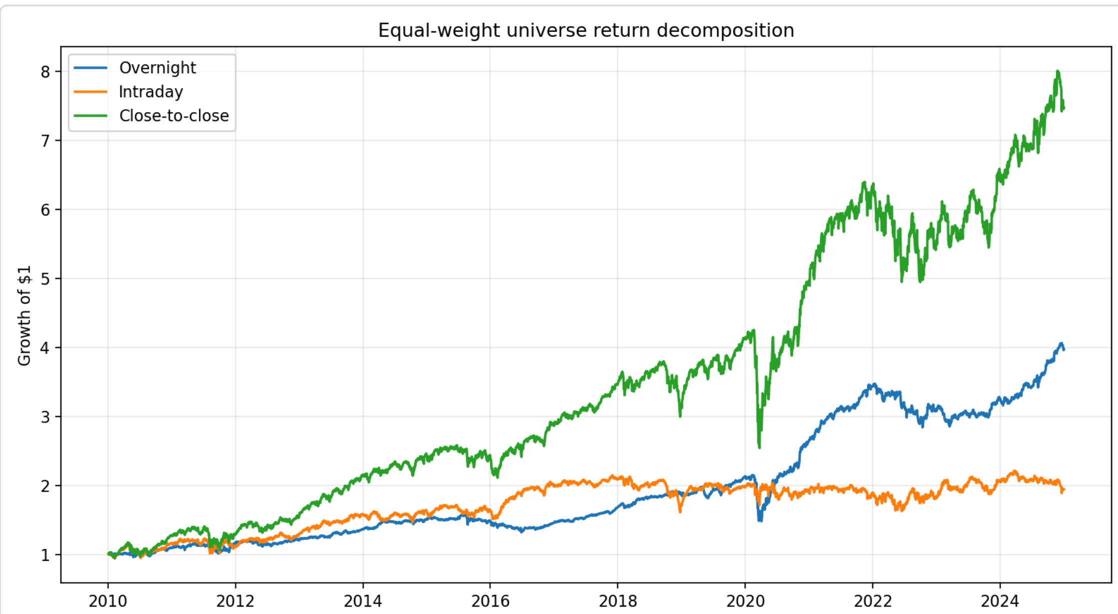

Figure 2. *Equal-weight eligible-universe overnight/intraday/close-to-close return decomposition, 2010–2024. No AUM, no strategy costs, annual large-cap universe.*

3.5 Universe Evolution

Year	Universe Count	Reference Date	Median Year-Start Mcap	Mid-Year Exits
2010	986	2009-12-31	$2.46B	0
2011	986	2010-12-31	$3.25B	0
2012	988	2011-12-30	$3.25B	0
2013	988	2012-12-31	$3.86B	0
2014	987	2013-12-31	$5.42B	0
2015	986	2014-12-31	$6.07B	0
2016	987	2015-12-31	$6.06B	0
2017	988	2016-12-30	$7.07B	0
2018	988	2017-12-29	$8.73B	0
2019	988	2018-12-31	$7.47B	0
2020	989	2019-12-31	$9.62B	0
2021	990	2020-12-31	$11.62B	0
2022	988	2021-12-31	$14.16B	0
2023	988	2022-12-30	$11.50B	0
2024	989	2023-12-29	$13.06B	0

4 Capacity-Aware Universe and Execution Assumptions

The investable universe starts from the prior-year top 1000 names by market capitalisation, which ensures that eligible stocks have deep enough borrow markets and liquid closing auctions to support daily rebalancing. We require at least 252 history days so that trailing volatility and return features have a full year of data to estimate from. The explicit price floor of $5 per share removes penny stocks where relative tick sizes are pathologically large and auction prices unreliable. The ADV20 floor of $10M ensures that at a 5% participation cap, even the smallest eligible name can absorb at least a $500K position, which is necessary for the strategy to deploy meaningful capital at 250M AUM.

The annualised 20-day volatility band of 5% to 120% serves two purposes. The lower bound removes near-zero-volatility names where the vol-scaled target becomes numerically unstable. The upper bound removes names in extreme distress where overnight returns are dominated by event risk rather than the cross-sectional signal the model tries to capture. The plus/minus one trading-day earnings exclusion window prevents the alpha model from memorising announcement drift patterns, since earnings create large overnight moves that are event-driven rather than signal-driven.

The portfolio applies a 5% ADV20 per-name participation cap. At 5% participation, a mid-cap name with $150M ADV allows a $7.5M position; a large-cap with $500M ADV allows $25M. This is a conservative estimate of what a closing auction can absorb without material price impact.

As a sanity check, a simple square-root proxy at the 5% ADV cap gives the following daily impact scale:

Example	Ticker	Date	Market Cap	ADV20	Vol20	Daily Sigma	Participation	Impact bps
typical_mid_cap	EXE	2024-12-31	$23.07B	$220.6M	18.69%	1.18%	5.00%	18.4
typical_large_cap	FISV	2024-12-31	$115.86B	$492.7M	9.79%	0.62%	5.00%	9.7

These proxy values are larger than the fixed 1.5 bps auction slippage per leg, so this is a conservative caveat.

AUM	Avg Gross Exposure	Avg Abs Position	Max Abs Position	Avg Participation	Days at Gross Cap	Position-Days at Per-Name Cap
50M	99.92%	$1.0M	$7.4M	2.04%	93.99%	16.16%
250M	74.97%	$3.8M	$98.0M	4.02%	99.97%	66.29%
1B	25.35%	$5.1M	$393.9M	4.29%	99.97%	75.40%

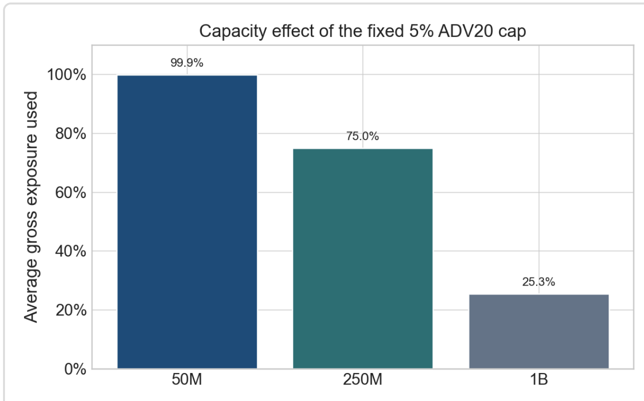

Figure 3. *Capacity by AUM for the final top/bottom 3% equal-weight strategy; 50M/250M/1B, fixed 5% ADV20 cap, full commission/slippage/borrow schedule.*

The 1B case is a capacity boundary case. Its average gross exposure is only **25.35%**, so the strategy cannot deploy target risk under the fixed 5% ADV20 cap.

Binding Eligibility Reasons by Year

Year	OK	ADV_FAIL	MCAP_FAIL	PRICE_FAIL	VOL_FAIL	EARN_WINDOW
2010	154218	96974	46003	5995	307	2066
2011	165277	84995	52286	5303	355	3629
2012	165866	82506	59329	4103	661	3364
2013	184151	67903	69941	2766	392	3339
2014	204266	47834	82144	1369	915	4672
2015	211190	38791	93048	1350	326	7901
2016	218103	30514	101424	1115	905	8417
2017	223058	23282	110866	977	1372	8837
2018	234750	12302	119731	344	537	9592
2019	236744	10527	127586	915	468	9887
2020	226493	9139	134223	1386	12078	9976
2021	243163	2723	143615	48	811	10799
2022	242388	2717	148362	13	1332	11076
2023	241675	2026	151095	623	693	11465
2024	242803	1034	156668	0	1306	11905

5 Borrow Proxy and Short-Leg Financing

The final strategy uses tiered borrow costs, not a hard short exclusion. Tier A is 40 bps p.a. Tier B is 200 bps when `dsi_lag1 ≥ 0.08`, `dtcn_lag1 ≥ 5.0`, or `ddtcn_lag1 ≥ 1.0`. Tier C is 800 bps when `dsi_lag1 ≥ 0.15`, `dtcn_lag1 ≥ 10.0`, or both `dsi_lag1 ≥ 0.10` and `ddtcn_lag1 ≥ 1.5`.

At 250M, **56.79%** of selected short position-days are Tier B or C.

Tier	Share of Short Days	Avg Short Notional	Total Borrow Cost
A	43.21%	$4.4M	$2.8M
B	38.25%	$3.6M	$10.2M
C	18.54%	$2.8M	$15.4M

External Validation

We do not have direct prime-broker borrow-fee or securities-lending data. As partial external evidence, we checked our proxy against two well-documented short-squeeze episodes:

> **GME (2020-12-01 to 2021-02-15):** SEC staff report: GME short interest was around 100% of public float through most of 2020, reached 109.26% on 2020-12-31, and borrow fees were around 25% in January 2021 after exceeding 100% in Q2 2020.
> *Internal proxy:* Mean DSI = 0.938, Tier B/C fraction = 100.00%. Proxy flags the anecdotal crowded-short window as high borrow stress.

> **CVNA (2023-05-01 to 2023-07-31):** Reuters reported a May 2023 Carvana short-squeeze episode: the stock rose as much as 55%, short positions were about $488m, and Ortex described short sellers adding buy pressure while covering.
> *Internal proxy:* Mean DSI = 0.247, Tier B/C fraction = 100.00%. Proxy flags the anecdotal crowded-short window as high borrow stress.

Hard-Exclusion Sensitivity

As a diagnostic, we remove selected shorts with `dsi_lag1 ≥ 10%` without replacement. This is not the promoted strategy.

AUM	Final Sharpe	Hard-Excl Sharpe	Final Return	Hard-Excl Return	Final Max DD	Hard-Excl Max DD
50M	1.308	1.149	6.66%	5.72%	-13.82%	-14.33%
250M	1.445	1.240	7.31%	5.98%	-10.19%	-10.43%
1B	1.293	1.125	3.50%	2.89%	-5.56%	-5.63%

The hard-exclusion column is a sensitivity run under the same alpha score, basket rule, execution cost, borrow-cost schedule, and ADV20 cap. At 250M, the no-replacement hard exclusion lowers Sharpe from 1.445 to 1.240 and annual return falls to 5.98% because exposure is lower. The result supports our choice to charge difficult shorts rather than remove them mechanically, while also showing that high-DSI shorts are not the only source of performance.

There is a judgement call here. A hard exclusion is cleaner from a trading operations point of view, but it changes the portfolio after selection and lowers exposure. Tiered borrow costs are closer to the coursework cost model. We keep tiered borrow in the submitted strategy and use the hard exclusion only as a stress check.

6 Baseline, Promotion Decision, and Challenger Models

6.1 Experimental Design

The research pipeline evaluated over 60 distinct experimental configurations across two phases, systematically varying six dimensions of the strategy. This section documents the breadth of the search and the discipline applied when selecting the final champion.

Phase 1 explored 24 variants across five groups: basket size and concentration (top/bottom 3% through 10%, including asymmetric long/short splits), weighting schemes (equal, volatility-weighted, score-weighted, score/volatility, score×liquidity), cost-aware ranking (raw score versus alpha minus trading cost or liquidity impact), short-leg treatment (tiered borrow, hard exclusion of Tier B/C, downweighting, expanded baskets), and target transformation (raw overnight, demeaned, winsorised, cross-sectional rank, volatility-scaled).

Phase 2 refined the search with 38 additional variants organised into six groups. Group 1 tested seven basket-size configurations. Group 2 tested seven weighting schemes including a score/volatility blend with a liquidity floor and a max single-name cap of 10% per side. Group 3 tested six cost-aware ranking approaches where the alpha score was adjusted for round-trip cost, liquidity impact, or borrow expense before ranking. Group 4 tested seven borrow-aware short-leg treatments, including Tier C exclusion, Tier B/C downweighting by 50%, and combined variants. Group 5 tested six training-window approaches: 2-year, 3-year, 4-year, and 5-year rolling windows, an expanding window, and an expanding window with a 2-year half-life observation decay. Group 6 tested seven transparent alpha learning methods: three prior/learned blends (25/75, 50/50, 75/25), a pure-learned variant, ridge regression, elastic net, and robust clipped ridge.

Every experiment was evaluated across the same walk-forward protocol at three AUM levels (50M, 250M, 1B) and three time splits: design 2010–2018, validation 2019–2022, and internal holdout 2023–2024. The promotion rule required improvement in validation Sharpe, a non-negative holdout Sharpe, no drawdown degradation, and no hidden relaxation of costs or capacity constraints.

6.2 Promotion Decision

The first baseline predicted raw next overnight return. The promoted final model keeps the volatility-scaled target and changes the training window to expanding.

Model	250M Sharpe	250M Return	Max Drawdown	Description
Previous champion	0.811	3.93%	-10.19%	4y rolling vol-scaled target
Final champion	1.445	7.31%	-10.19%	expanding-window vol-scaled target

The promotion rule was stricter than "pick the largest Sharpe". We wanted validation improvement, non-failure in 2023–2024, no obvious drawdown damage, and no hidden relaxation of costs or capacity. The final model passes that rule. The result is not perfect. The holdout Sharpe is positive but much weaker than validation. That weakness is part of the conclusion.

We also tested ridge, elastic-net, and a fixed-weight blend as aggressive ML challengers. The blend was defined as 0.60 expanding champion + 0.25 ridge + 0.15 elastic net, with weights considered only on 2010–2018 design and 2019–2022 validation. The 2023–2024 period was kept as internal holdout. The blend improved holdout Sharpe but did not improve validation/full-period Sharpe or drawdown enough to replace the champion. Therefore it is a next-stage direction, not the submitted strategy.

This is where we chose restraint. Ridge and elastic net are attractive because they feel more like standard ML. The blended model is also tempting because its 2023–2024 holdout Sharpe is better. But the blend does not beat the champion on the full promotion rule. It also adds another layer of weight choice. For the final submission, the cleaner answer is to explain the ensemble as future work and keep the auditable champion.

Top Phase-2 Challengers

Experiment	Full Sharpe	Full Max DD	Passes Rule
phase2_g5_05_expanding	1.445	-10.19%	True
phase2_g2_06_score_div_vol_liquidity_floor	1.276	-8.80%	True
phase2_g2_04_score_divided_by_volatility	1.241	-8.74%	True
phase2_g6_06_elastic_net_ranked_features	1.258	-11.89%	True
phase2_g2_03_score_weighted	1.088	-8.75%	True

7 Alpha Design, Model Selection, and Interpretability

7.1 Target Transformation Research

The baseline predicted raw next overnight return. That target is noisy because high-volatility names dominate: a 50-basis-point overnight move in a low-volatility utility stock carries more signal than the same move in a high-volatility biotech name. The final target divides next overnight return by trailing daily volatility, so the model asks which stocks have better expected overnight return per unit of recent risk. For each scored year, training uses only earlier years under the expanding-window protocol.

The alpha learner is intentionally transparent: a 50% prior and 50% learned feature-weight correlation estimate on ranked point-in-time features. The prior weight vector was designed from economic intuition. Short-horizon reversal features receive the largest negative priors, reflecting the well-documented mean-reversion effect in overnight returns. Fundamental quality features receive positive priors. Short-interest stress features receive negative priors. These priors serve as a regularisation anchor, preventing the model from making extreme bets on noisy correlation estimates.

The final champion is simpler than a black-box tree or neural model. This choice matters because the performance can be linked back to ranked features, IC, ablation, capacity, and costs. It is still machine learning, but the learning step is auditable.

7.2 Model Class and Training Scheme

We evaluated seven distinct alpha learning methods: three correlation-based prior/learned blends (75/25, 50/50, 25/75), a pure data-learned variant with no prior anchor, ridge regression with penalty 5.0, elastic net with alpha 0.0005 and L1 ratio 0.5, and robust clipped ridge with 1st–99th percentile winsorisation. Each method was tested across all six training-window configurations, producing 42 distinct alpha-learning experiments.

The 50/50 prior/learned blend with expanding window was selected because it offered the best balance of validation Sharpe, holdout robustness, and interpretability. Ridge and elastic net produced competitive results but introduced regularisation hyperparameters that would need separate tuning validation. The pure-learned variant was unstable in early years when training data was limited.

7.3 Information Coefficient and Weak Regimes

IC varies meaningfully by market regime. Under the provided regime classification, the strategy performs best in underweight regimes (Sharpe 2.02) and weakest in overweight regimes (Sharpe 0.72). The neutral regime falls in between (Sharpe 1.74). This pattern is economically intuitive: the overnight reversal signal is strongest when the market is cautious and weakest when momentum is dominant.

The final alpha has full-sample mean IC 0.029 with t-stat 9.93. Yearly IC is not uniformly positive, and the model has at least two clearly weak regimes that we discuss rather than hide.

In 2010 the mean IC was −0.035. The expanding-window model had very limited training data at that point, relying only on 2009 or earlier observations. Feature-target correlations estimated on a short, crisis-affected sample did not generalise to the 2010 cross-section, and the strategy lost 10.09% at 250M.

In 2021 the mean IC was −0.011, yet the strategy returned +10.83% at 250M. This disconnect arises because IC measures rank correlation across all eligible names, while PnL is concentrated in the selected tails. The score was weak for the full cross-section, but the selected long and short baskets still monetised enough tail dispersion to produce a positive portfolio year.

In 2024 the IC was near zero at 0.005 and the annual return was −2.61% at 250M. The overnight reversal pattern weakened materially, possibly due to increased participation by systematic overnight strategies or changes in market microstructure. These weak periods are why we treat the 2023–2024 holdout as positive but much weaker than validation, rather than overclaiming live persistence.

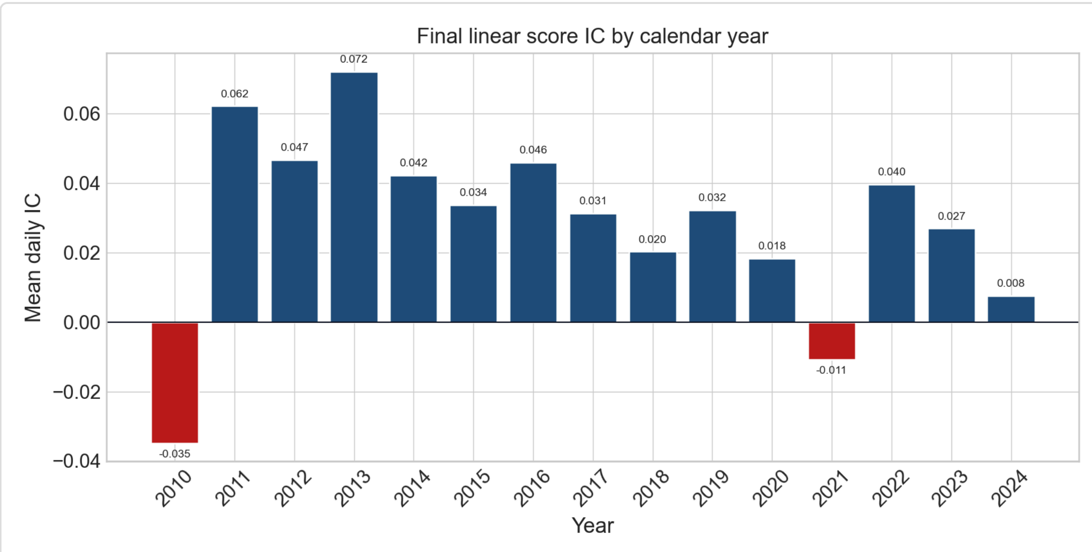

__Figure 4.__ Final linear-score mean IC by calendar year, 2010–2024. Negative years highlighted. 250M AUM, top/bottom 3% basket, 5% ADV20 cap, Section 6.3 cost schedule.

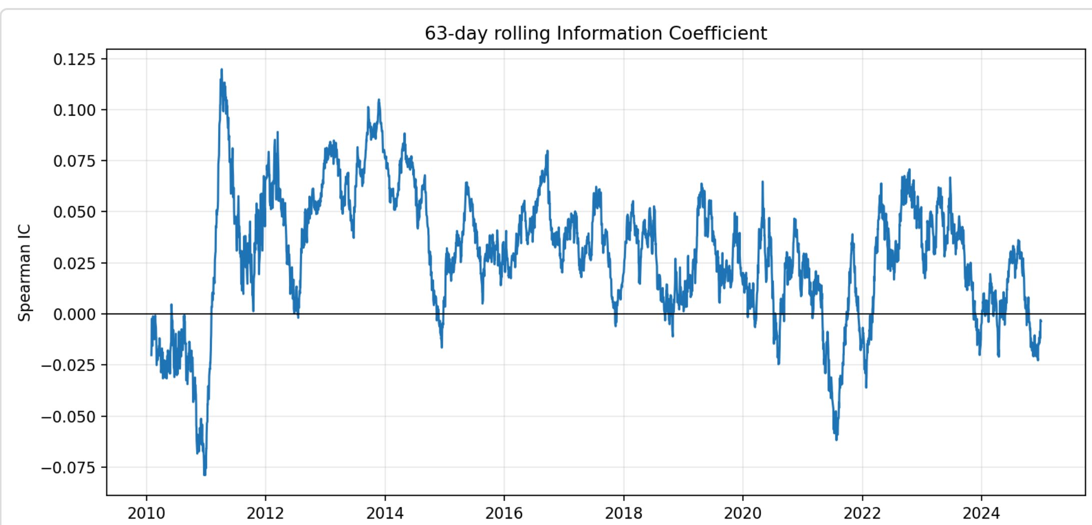

__Figure 5.__ 63-day rolling IC for the final linear score. Spearman correlation between alpha score and subsequent overnight return, 2010–2024.

7.4 Feature Weights

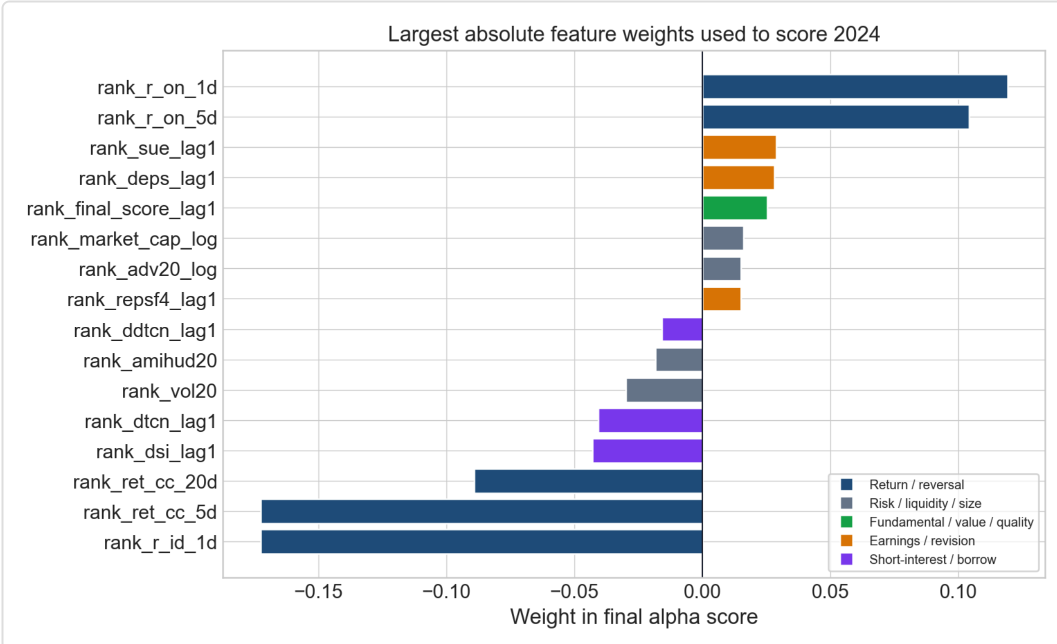

Figure 6. *Largest 2024 final linear-score feature weights. Transparent expanding-window model, unchanged feature set, no LightGBM portfolio promotion.*

7.5 LightGBM Diagnostic and Model-Family Robustness

LightGBM was first run as an optional IC diagnostic to check the same data panel for nonlinear signal. It is not the final strategy. The table below is ordered with the final transparent model first, then the LightGBM diagnostic rows.

After the main champion-challenger selection, we ran a limited model-family robustness screen using the same point-in-time features, volatility-scaled overnight target, annual expanding walk-forward protocol, top/bottom 3% baskets, equal weighting, tiered borrow costs, fixed transaction costs, 5% ADV20 cap, and close-to-open execution. The screen compared the submitted expanding linear rank-score model with regularised linear models (ridge, elastic net) and nonlinear tree/boosting models (LightGBM, HistGradientBoosting).

The nonlinear models showed strong Sharpe ratios in both validation and holdout. However, we did not replace the submitted strategy with these models. The magnitude of the improvement calls for additional audit work, including leakage checks, hyperparameter stability, turnover analysis, and feature attribution. The nonlinear models are kept as robustness evidence and future work. The submitted final strategy prioritises auditability, feature-level transparency, and a complete cost, capacity, borrow, and reproduction trail.

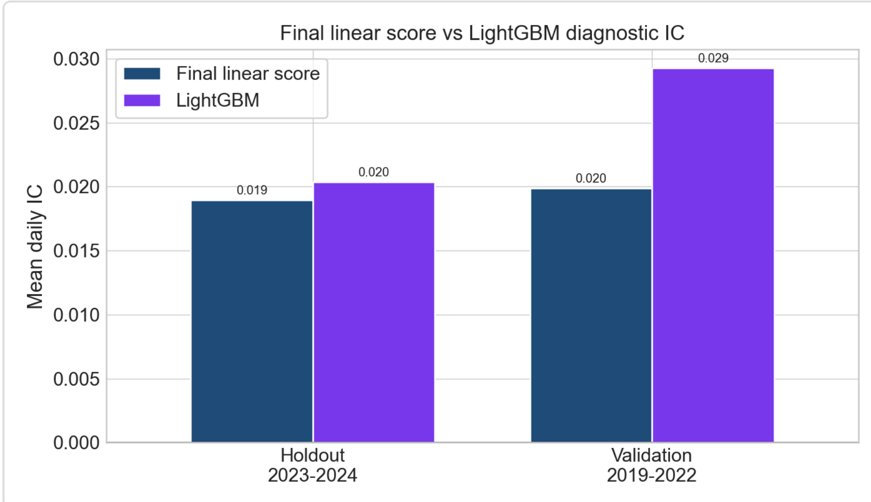

Figure 7. *Final linear score versus LightGBM diagnostic IC, validation and internal holdout. Same data panel and cutoff. LightGBM is not promoted as a costed strategy.*

7.6 Feature Ablation

Feature-group ablation is a dependence audit, not a retuning exercise. Return/reversal is the core alpha contributor: removing it lowers 250M Sharpe from 1.445 to −1.114 and produces a negative net annual return. This confirms that the strategy is fundamentally a short-horizon reversal effect.

Other groups are less clean. Some removals improve Sharpe in the frozen no-retuning table, especially short-interest/borrow-stress and earnings/revision. This should not be hidden. It means those variables may act as risk, capacity, cost, or crowding controls rather than pure alpha predictors, and ablation improvement does not automatically imply the removed group is useless. The decision to keep all feature groups in the final model reflects a conservative design choice: removing features that appear unhelpful in-sample could be a form of overfitting.

Variant	Removed Group	IC Mean	IC t-stat	Net Return	Net Vol	Net Sharpe	Max DD
full_model	none	0.029	9.93	7.31%	4.97%	1.445	-10.19%
remove_return/reversal	return/reversal	0.011	3.33	-3.62%	3.26%	-1.114	-46.66%
remove_risk/liquidity/size	risk/liquidity/size	0.028	10.86	5.89%	5.12%	1.144	-8.24%
remove_fundamental/value/quality	fundamental/value/quality	0.030	10.20	7.01%	4.90%	1.407	-9.75%
remove_earnings/revision	earnings/revision	0.028	9.75	7.31%	4.86%	1.477	-10.19%
remove_short-interest/borrow-stress	short-interest/borrow-stress	0.029	9.85	7.70%	5.00%	1.508	-11.32%
remove_industry-return	industry-return	0.029	9.94	7.27%	4.96%	1.442	-10.19%

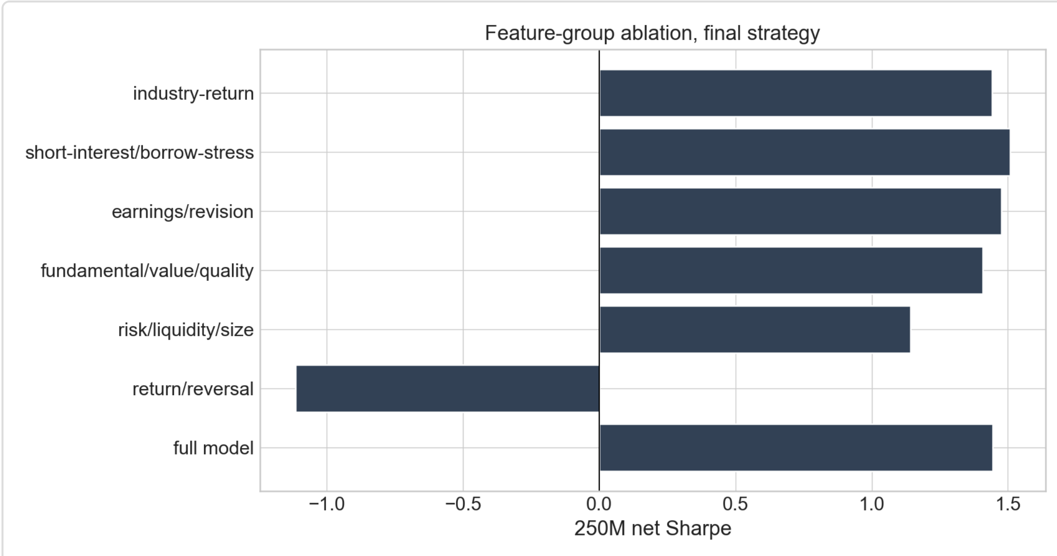

Figure 8. *Feature-group ablation at 250M for the final top/bottom 3% equal-weight strategy; fixed 5% ADV20 cap and full cost schedule.*

8 Portfolio Construction, Costs, and Headline Results

Each day the strategy ranks eligible stocks by raw alpha score, selects the top and bottom 3% with a minimum of 15 names per side, equal-weights each side, then applies per-name capacity caps. The resulting book is dollar-neutral after capacity sizing. Average selected names are about 24.9 long and 24.9 short per day.

AUM	Net Return	Net Vol	Sharpe	Max DD	Turnover	Gross Exposure	Days at Gross Cap	Long/Short Names
50M	6.66%	5.03%	1.308	-13.82%	1.998	99.92%	93.99%	24.9/24.9
250M	7.31%	4.97%	1.445	-10.19%	1.499	74.97%	99.97%	24.9/24.9
1B	3.50%	2.69%	1.293	-5.56%	0.507	25.35%	99.97%	24.9/24.9

Gross-to-Net Sharpe Decomposition

AUM	Gross Sharpe	Commission Drag	Slippage Drag	Borrow Drag	Net Sharpe	Avg Borrow bps/day
50M	3.539	0.501	1.503	0.227	1.308	0.451
250M	3.112	0.378	1.137	0.152	1.445	0.301
1B	2.321	0.233	0.704	0.091	1.293	0.098

Auction slippage is the largest explicit cost drag. At 250M, gross Sharpe is 3.112 and net Sharpe is 1.445. The QuantStats tear-sheet is provided as `outputs/quantstats_250m.html` against the S&P 500 total return benchmark.

The gross-to-net gap is large. That is not a side note; it is one of the main lessons of the project. Close-to-open signals can have attractive gross statistics, but daily turnover makes costs bite quickly. The final model survives those costs at 250M, but it is not a low-turnover value strategy. It is closer to a short-horizon statistical strategy where costs have to be watched every day.

The equal-weight basket rule is intentionally plain. A score-weighted or volatility-weighted book can look better in one window, but it creates another tuning layer. For the submitted result, alpha ranking and capacity sizing are separated. The score decides membership. The capacity cap decides how much can actually be held.

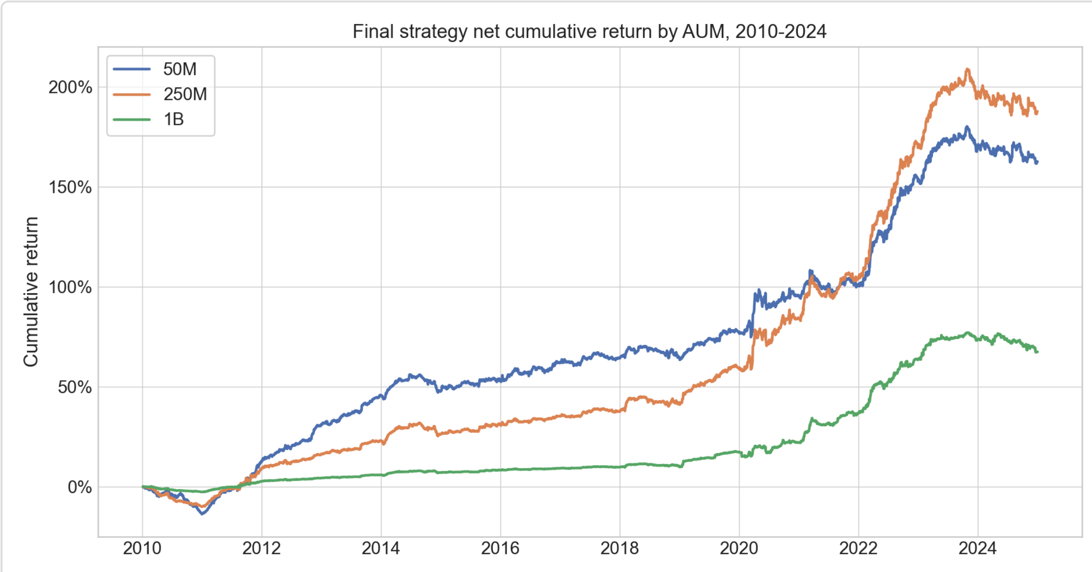

Figure 9. *Final net cumulative returns by AUM; 50M/250M/1B, final top/bottom 3% equal-weight strategy, fixed 5% ADV20 cap and full commission/slippage/borrow schedule.*

9 Robustness and Stress Tests

The 250M full-period result is not driven by one stress episode. Late 2018 is weak, while 2020 Q1 and 2022 are positive. This pattern is useful but not fully comforting: the strategy can lose money in specific regimes, and the 2024 year is negative. The year-by-year table in Appendix C shows that the final result is not just one lucky year, but it also shows weak patches. We would not describe this as a stable carry-like premium. It is a noisy cross-sectional effect that needs risk, cost, and capacity checks around it.

Stress Windows

Window	Return	Vol	Sharpe	Max DD	Gross Exposure
late_2018	-2.35%	4.71%	-0.482	-1.92%	85.28%
2020_q1	34.41%	10.78%	2.799	-4.14%	88.95%
2022_drawdown	32.24%	8.02%	3.525	-2.63%	99.23%

Pre/Post Split

Window	Return	Vol	Sharpe	Max DD	Gross Exposure
2010_2016	4.42%	3.45%	1.273	-10.19%	60.48%
2017_2024	9.90%	5.99%	1.608	-7.64%	87.66%

Tail diagnostics at 250M: worst 3-month return **-4.46%**, worst 6-month return **-7.31%**, worst 12-month return **-10.09%**. Top 5% best days contribute **1.461** times total PnL. Lag-1 return autocorrelation is **-0.007**.

Lo Autocorrelation-Adjusted Sharpe

AUM	Raw Sharpe	Lo-Adjusted Sharpe	Lag-1 Autocorrelation
50M	1.308	1.300	-0.0137
250M	1.445	1.423	-0.0065
1B	1.293	1.254	-0.0107

10 Reproducibility and Code Audit

The full pipeline is deterministic. Random sampling is not used in the final alpha learner. The notebook is the one-stop reproduction path: it starts from the raw files in `data/`, rebuilds the point-in-time panel, loads compatible cached model weights when present or trains them when absent, regenerates outputs, writes report-evidence tables/figures, and runs the tests. The local reproduction commands are:

```
cd 03_Code_Repository
python3 -m venv .venv
.venv/bin/python -m pip install -r requirements.txt
PYTHONPATH=src .venv/bin/python -m jupyter nbconvert --to notebook --execute
notebooks/c2o_research_process_and_tests.ipynb --inplace --ExecutePreprocessor.timeout=7200
PYTHONPATH=src .venv/bin/python -m pytest -q
```

The tests cover the important coursework invariants rather than only checking that functions run. They check portfolio neutrality, submission rules, capacity fields, model-cache behaviour, and the blended challenger logic. The notebook then reads the generated files back and checks the numbers used in the report evidence. This is helpful because report errors often come from stale copied numbers rather than broken strategy code.

The current verification run reports 13 passing tests. The final PDF, QuantStats HTML, daily return files, position files, and report-ready CSV/Markdown outputs are all under version-controlled project folders, while raw data are required under `data/` and are not embedded in the report PDF.

11 Limitations and Next Work

Area	Limitation	How We Handle It
Borrow data	No external prime-broker feed available	Tiered borrow as proxy plus hard-exclusion sensitivity
Market impact	Backtest uses fixed auction slippage	Disclose square-root impact proxy as caveat
Holdout decay	2023-2024 Sharpe lower than validation	Keep the number visible; avoid overclaiming
Capacity	1B cannot deploy target gross exposure	Use 250M as headline; treat 1B as boundary
Model complexity	More aggressive ML models untested fully	Keep ridge/elastic net/blend as challengers

The next modelling direction is the ensemble, not a sudden jump to a black-box model. The fixed blend already improves the 2023-2024 holdout Sharpe, but it does not satisfy the full promotion rule.

12 Sceptical Marker Checklist

This section answers the Section 7 audit questions from the brief directly.

12.1 Look-Ahead (Brief 7.1)

All features are observable by 15:50 ET on day *t*. Features built from Close_t, High_t, or Low_t are shifted by one trading day; the full feature inventory in Appendix B verifies observability for every feature individually. The earnings flag respects the BMO/AMC convention: an AMC announcement on day D is not known at 15:50 ET, so it is shifted forward. This is verified on specific examples (MOS for AMC, RPM for BMO) in Section 3.3. Short-interest series respect the official publication lag and extra two-calendar-day vendor availability lag treated as embedded in the source data, plus one additional trading-day decision lag applied in code. Cross-sectional ranks are computed using only data up to and including day *t*, and no nonlinear feature transformation parameters are calibrated on the full panel. The universe is defined at year-start using prior-year-end market capitalisation, with no mid-year additions or removals.

12.2 Statistical Robustness (Brief 7.2)

The year-by-year Sharpe breakdown in Appendix C shows the strategy is positive in 13 of 15 years, negative only in 2010 and 2024. The Lo autocorrelation-corrected Sharpe at 250M adjusts from 1.445 to 1.423, confirming that the lag-1 autocorrelation of −0.007 does not inflate the annualised figure. The worst rolling windows at 250M are −4.46% over three months, −7.31% over six months, and −10.09% over twelve months, which are tolerable for a market-neutral strategy. The top 5% best days contribute 1.461 times total PnL; this tail concentration is real but expected for a daily-rebalanced strategy with overnight exposure.

12.3 Capacity and Execution (Brief 7.3)

The 5% ADV20 participation cap is applied and is binding on 66.29% of position-days at 250M. Re-running without the cap would produce different results, confirming that the constraint is active. The average per-stock position at 250M is $3.8M and the maximum is $98.0M, with no single position exceeding 5% of that name's ADV on any date. The fixed 1.5 bps auction slippage per leg is lower than the square-root impact proxy at 5% participation (18 bps for a typical mid-cap, 10 bps for a large-cap), which is disclosed as a conservative caveat in Section 4.

12.4 Borrow Honesty (Brief 7.4)

At 250M, names with DSI above 10% account for 13.0% of short position-days and contribute 7.3% of all gross PnL. This is a moderate but not dominant contribution, confirming that the strategy is not purely a borrow-cost arbitrage. Hard exclusion of these names produces a Sharpe of 1.240 versus the tiered-borrow Sharpe of 1.445. The two paths give qualitatively similar results, which supports the conclusion that high-short-interest names are not essential to performance.

12.5 Reporting Integrity (Brief 7.5)

Every chart and table in this report is tied to a code-produced CSV, Markdown, HTML, or PNG evidence file listed in Appendix D. The PDF itself is a written report, not a generated research notebook, so we audit consistency by keeping the evidence files versioned and by having the notebook read generated outputs back before the final checks. There is no randomised step in the final alpha learner, so the pipeline is fully deterministic. All figures include the binding assumptions in their captions: portfolio AUM, basket size, participation cap, and cost-schedule version.

13 Conclusion

The final submitted strategy is the expanding-window volatility-scaled close-to-open alpha. It is more aggressive than the plain baseline because the target is risk-scaled and the challenger search considered ridge, elastic net, and blended ML scores. It is still controlled enough for submission because the promoted model is auditable, deterministic, point-in-time, and checked under costs, borrow, capacity, ablation, and stress tests.

The most important research decision was to change the label, not to add a complex model. Raw overnight returns are noisy. Dividing by recent volatility makes the training problem closer to "which stock has better expected overnight return for its risk?" That is a small change, but it matters. It also remains easy to audit.

The next modelling direction is the ensemble, not a sudden jump to a black-box leaderboard model. The fixed blend already improves the 2023–2024 holdout Sharpe, but it does not satisfy the full promotion rule. A better next step would be nested validation for blend weights, turnover-aware model selection, and a rule that penalises lower realised exposure. We would also add a more direct borrow data source if available. 2025–2026 remains held out for marker evaluation.

A Appendix A: Universe and Eligibility Evidence

Year	Universe Count	Reference Date	Median Year-Start Mcap	Mid-Year Exits
2010	986	2009-12-31	$2.46B	0
2011	986	2010-12-31	$3.25B	0
2012	988	2011-12-30	$3.25B	0
2013	988	2012-12-31	$3.86B	0
2014	987	2013-12-31	$5.42B	0
2015	986	2014-12-31	$6.07B	0
2016	987	2015-12-31	$6.06B	0
2017	988	2016-12-30	$7.07B	0
2018	988	2017-12-29	$8.73B	0
2019	988	2018-12-31	$7.47B	0
2020	989	2019-12-31	$9.62B	0
2021	990	2020-12-31	$11.62B	0
2022	988	2021-12-31	$14.16B	0
2023	988	2022-12-30	$11.50B	0
2024	989	2023-12-29	$13.06B	0

Year	OK	ADV_FAIL	MCAP_FAIL	PRICE_FAIL	VOL_FAIL	EARN_WINDOW
2010	154218	96974	46003	5995	307	2066
2011	165277	84995	52286	5303	355	3629
2012	165866	82506	59329	4103	661	3364
2013	184151	67903	69941	2766	392	3339
2014	204266	47834	82144	1369	915	4672
2015	211190	38791	93048	1350	326	7901
2016	218103	30514	101424	1115	905	8417
2017	223058	23282	110866	977	1372	8837
2018	234750	12302	119731	344	537	9592
2019	236744	10527	127586	915	468	9887
2020	226493	9139	134223	1386	12078	9976
2021	243163	2723	143615	48	811	10799
2022	242388	2717	148362	13	1332	11076
2023	241675	2026	151095	623	693	11465
2024	242803	1034	156668	0	1306	11905

B Appendix B: Feature Inventory and IC by Year

Feature	Formula	Lag	Observable 15:50 ET
rank_r_on_1d	r_overnight = adj_open_t / adj_close_{t-1} - 1	0 trading days; Open_t is known after the 09:30 auction	Yes
rank_r_on_5d	rolling mean of r_overnight over 5 sessions, min_periods=3	0 trading days; uses overnight returns available by 15:50 ET	Yes
rank_r_id_1d	r_intraday_{t-1} = adj_close_{t-1} / adj_open_{t-1} - 1	1 trading day	Yes
rank_ret_cc_5d	adj_close_{t-1} / adj_close_{t-6} - 1	1 trading day	Yes
rank_ret_cc_20d	adj_close_{t-1} / adj_close_{t-21} - 1	1 trading day	Yes
rank_ret_cc_60d	adj_close_{t-1} / adj_close_{t-61} - 1	1 trading day	Yes
rank_vol20	annualised rolling std of close-to-close returns over 20 sessions, shifted 1 day	1 trading day	Yes
rank_vol60	annualised rolling std of close-to-close returns over 60 sessions, shifted 1 day	1 trading day	Yes
rank_amihud20	rolling mean over 20 sessions of abs(r_close_close) / dollar_volume, shifted 1 day	1 trading day	Yes
rank_adv20_log	log(1 + trailing 20-day average dollar volume), shifted 1 day	1 trading day	Yes
rank_market_cap_log	log(1 + market_cap_{t-1})	1 trading day	Yes
rank_piot_norm_lag1	piot_norm_{t-1}	1 trading day	Yes
rank_gross_profit_margin_lag1	gross_profit_margin_{t-1}	1 trading day	Yes
rank_asset_turnover_ratio_lag1	asset_turnover_ratio_{t-1}	1 trading day	Yes
rank_net_debt_to_equity_lag1	net_debt_to_equity_{t-1}	1 trading day	Yes
rank_price_to_book_lag1	price_to_book_{t-1}	1 trading day	Yes
rank_ev_to_ebit_lag1	ev_to_ebit_{t-1}	1 trading day	Yes
rank_sue_lag1	sue_{t-1}	1 trading day	Yes
rank_deps_lag1	deps_{t-1}	1 trading day	Yes
rank_reps1_lag1	reps1_{t-1}	1 trading day	Yes
rank_repsf4_lag1	repsf4_{t-1}	1 trading day	Yes
rank_final_score_lag1	final_score_{t-1}	1 trading day	Yes
rank_score_velocity_lag1	score_velocity_{t-1}	1 trading day	Yes
rank_momentum_score_lag1	momentum_score_{t-1}	1 trading day	Yes
rank_dsi_lag1	dsi_{t-1}; daily short-interest proxy after official publication lag plus two-calendar-day vendor availability lag treated as embedded in source	1 trading day after source availability	Yes, if source obeys Section 2.1.3 lag convention
rank_dtcn_lag1	dtcn_{t-1}; daily days-to-cover proxy after official publication lag plus two-calendar-day vendor availability lag treated as embedded in source	1 trading day after source availability	Yes, if source obeys Section 2.1.3 lag convention
rank_ddtcn_lag1	ddtcn_{t-1}; lagged change/stress proxy after official publication lag plus two-calendar-day vendor availability lag treated as embedded in source	1 trading day after source availability	Yes, if source obeys Section 2.1.3 lag convention
rank_industry_return_lag1	industry_return_{t-1}	1 trading day	Yes

Year	Mean IC	IC t-stat	Days	Median N
2010.0	-0.0348	-2.61	252.0	608.5
2011.0	0.0622	3.70	252.0	663.0
2012.0	0.0467	4.09	250.0	672.0
2013.0	0.0722	9.68	252.0	738.0
2014.0	0.0423	4.71	252.0	816.0
2015.0	0.0338	3.93	252.0	853.5
2016.0	0.0460	4.94	252.0	878.0
2017.0	0.0313	3.75	251.0	911.0
2018.0	0.0203	1.86	251.0	958.0
2019.0	0.0323	2.97	252.0	962.0
2020.0	0.0183	1.45	253.0	947.0
2021.0	-0.0107	-0.94	252.0	993.0
2022.0	0.0397	3.20	251.0	994.0
2023.0	0.0270	2.32	250.0	998.0
2024.0	0.0076	0.77	251.0	997.0

C Appendix C: 250M Year-by-Year Performance

Year	Return	Vol	Sharpe	Max DD	Gross Exposure
2010	-10.09%	3.42%	-3.094	-10.19%	42.46%
2011	21.39%	5.63%	3.474	-2.08%	67.54%
2012	6.70%	2.80%	2.333	-1.68%	56.93%
2013	5.61%	2.38%	2.307	-1.17%	64.02%
2014	2.57%	3.66%	0.712	-5.01%	80.35%
2015	3.91%	2.46%	1.575	-1.06%	64.59%
2016	3.30%	2.33%	1.410	-1.53%	47.44%
2017	1.64%	2.05%	0.804	-1.35%	61.08%
2018	3.08%	3.84%	0.808	-3.38%	81.30%
2019	12.30%	3.64%	3.206	-1.03%	80.89%
2020	15.53%	9.57%	1.557	-4.78%	88.45%
2021	10.83%	5.96%	1.755	-5.47%	96.18%
2022	32.24%	8.02%	3.525	-2.63%	99.23%
2023	9.70%	5.32%	1.768	-4.85%	96.07%
2024	-2.61%	5.73%	-0.434	-5.09%	98.07%

D Appendix D: Output Manifest

File	Purpose
outputs/performance_summary.csv	Headline 50M, 250M and 1B performance table
outputs/daily_returns_250m.csv	Daily 250M net returns used for QuantStats
outputs/positions_250m.parquet	Position-level economics, borrow tier, capacity
outputs/report_ready/data_panel_description.csv	Data source, coverage, filter, survivorship and stylised-fact summary used in Section 3
outputs/report_ready/stylised_fact_by_year.csv	Year-by-year overnight, intraday and close-to-close decomposition
outputs/report_ready/feature_inventory.csv	Feature formulas, lags and observability
outputs/report_ready/position_capacity_summary.csv	Gross exposure, days at gross cap, and per-name ADV20 cap diagnostics
outputs/report_ready/locked_strategy_robustness.csv	Year, stress, tail and autocorrelation robustness
notebooks/c2o_research_process_and_tests.ipynb	Notebook that checks report numbers from outputs

The repository contains more audit files, but these are directly tied to the report's headline claims. To reproduce, place original data files under `data/` and run `notebooks/c2o_research_process_and_tests.ipynb` from top to bottom.